

# NextGenGrowth.in Tech Stack & Campus Landing Page Recommendation Guide

Comprehensive technical overview of the current NextGenGrowth platform and strategic recommendations for building a high-converting Campus Marketing Landing Page.

Platform: **NextGenGrowth.in**  
(v3.0)

Target  
Audience:

**College Students, Ambassadors & Brand  
Partners**

Date: **August  
2026**

## 01 Executive Summary / Team Context

### टीम के लिए सन्देश (Context):

यह डॉक्यूमेंट NextGenGrowth platform (nextgengrowth.in) के पुरे Tech Stack को explain करता है और Campus Marketing Landing Page बना रही टीम के लिए best technical recommendations प्रदान करता है। इसका मुख्य उद्देश्य Campus Landing Page को extremely fast, mobile-friendly, scalable और NextGenGrowth backend API के साथ smoothly connect करना है।

## 02 Full Tech Stack Architecture (Current Platform)

### Frontend Layer

Client Side

- ▶ **Framework:** React 18.3 (Component-driven SPA)
- ▶ **Build Tool:** Vite 5.2 (Ultra-fast HMR & bundling)
- ▶ **Routing:** React Router DOM v6.22
- ▶ **Icons & Graphics:** Lucide React (Modern, light iconography)
- ▶ **Animations:** Framer Motion 12 + GSAP 3.15
- ▶ **Smooth Scroll Engine:** Lenis Smooth Scroll 1.3
- ▶ **3D Visual Elements:** Three.js 0.184 (Interactive 3D canvas)
- ▶ **State & Context:** React AuthContext + Custom Hooks

### Backend & API Engine

Server Side

- ▶ **Runtime:** Node.js (v18+)
- ▶ **Web Framework:** Express.js 4.22
- ▶ **Authentication:** JWT (JSON Web Tokens) + Bcryptjs
- ▶ **OAuth 2.0 Integration:** Passport.js + Google OAuth 2.0
- ▶ **Session & Security:** Express-Session, Express-Rate-Limit
- ▶ **CORS & Security:** CORS origin protection, Dotenv config

### Database & Persistence

Data Layer

- ▶ **Primary Database:** MongoDB Atlas (NoSQL Document Store)
- ▶ **Object Data Modeling (ODM):** Mongoose 8.2
- ▶ **Core Schemas:** User (Students & Brands), Projects, Applications, Workspaces, KYC Credentials, Ambassador Leads
- ▶ **Data Security:** Password hashing, Encrypted Tokens

### AI, Payments & Integrations

Services

- ▶ **AI Engine:** Google Gemini AI (@google/generative-ai) for AI brief generator & talent matching
- ▶ **Payment Escrow:** Razorpay Node.js SDK 2.9 (Order creation & Webhook validation)
- ▶ **Email & Outreach:** Resend API 6.9 + Nodemailer 9.0
- ▶ **Utilities:** ExcelJS, QRCode generation, Playwright E2E

### Architecture Quick Reference Table

| LAYER       | TECHNOLOGY SELECTED            | KEY RESPONSIBILITY / FUNCTION                                                             |
|-------------|--------------------------------|-------------------------------------------------------------------------------------------|
| Client UI   | React 18 + Vite + Tailwind/CSS | Delivers sub-second SPA loading, dark-themed futuristic UX, interactive micro-animations. |
| Backend API | Node.js + Express.js           | Handles Auth, User Verification, Student/Brand Dashboards, Project Postings, Escrow.      |
| Database    | MongoDB Atlas + Mongoose       | Stores profiles, campus ambassador leads, project deliverables, and transaction history.  |
| AI Engine   | Google Gemini 1.5 / Flash      | Powers instant project brief creation and automatic student skill matching.               |
| Payments    | Razorpay Escrow API            | Secure milestone payments between Brands and Student Freelancers/Ambassadors.             |
| Outreach    | Resend API + Nodemailer        | Automated welcome emails, campus campaign updates, and application status alerts.         |

## 03 Recommendations for New Campus Marketing Landing Page

क्योंकि नया Landing Page मुख्य रूप से **Campus Marketing & Student Ambassador Acquisition** के लिए है, इसलिए इसे निम्नलिखित मापदंडों (Criteria) पर पूरा उतरना चाहिए:

### ★ Recommended Tech Stack Approach for Team

#### Option 1 (Best Choice for Speed & Existing Component Reuse): React 18 + Vite + Tailwind CSS + Framer Motion + Lenis

यह NextGenGrowth के मौजूदा Client codebase के साथ 100% compatible है। Team existing Navbar, Footer, Auroras, Hero Sections और Modals को आसानी से reuse कर सकती है।

#### Option 2 (Alternative if SEO & Pure Static Performance is Highest Priority): Astro 4.0 or Next.js 14 (App Router)

अगर टीम Static Site Generation (SSG), 99+ Lighthouse performance score और zero-JS initial load चाहती है तो Astro बेस्ट है।

### 1. UI/UX & Interactive Design

- ▶ **Dark Aurora Aesthetics:** Dark theme (#0F172A) with glassmorphism cards and neon accent gradients (#3B82F6, #06B6D4).
- ▶ **Smooth Scrolling:** Integrate Lenis smooth scroll for effortless scrolling experience across mobile & desktop.
- ▶ **Interactive Widgets:** Include Campus Stipend/Earnings Calculator (similar to CampusCalculator.jsx) and Ambassador Ecosystem Flywheel.

### 2. Mobile-First Optimization

- ▶ **Mobile Priority:** 85%+ students target link through Instagram/WhatsApp. Layouts must be 100% responsive.
- ▶ **Fast Loading:** Use WebP image formats, font-display: swap, and lazy loading for heavy animations.
- ▶ **Clickable CTAs:** Instant sticky "Apply as Ambassador" button on mobile screens.

### 3. API Handshake & Lead Generation

- ▶ **Direct API Hook:** Connect form submissions directly to NextGenGrowth server API: POST /api/ambassadors/register or POST /api/campus/lead.
- ▶ **Auto-Response Email:** Trigger instant confirmation email via Resend API upon student form submission.
- ▶ **Spam Protection:** Use Cloudflare Turnstile or Google reCAPTCHA v3 on signup forms.

### 4. Analytics & Conversion Tracking

- ▶ **Meta Pixel & LinkedIn Insight:** Track conversions for Instagram/LinkedIn ad campaigns.
- ▶ **Google Analytics 4 (GA4):** Funnel tracking from Hero section click to Ambassador modal submission.
- ▶ **UTM Tracking:** Pass UTM parameters (source, campus, ambassador code) into lead database.

04 API Integration Blueprint (Code Snippet for Team)

Sample reusable integration function for the team to send Campus Ambassador signup leads straight to NextGenGrowth API:

```
// Campus Lead Form Submission Handler async function handleCampusSignup(formData) { try { const response = await fetch('https://www.nextgengrowth.in/api/ambassadors/register', { method: 'POST', headers: { 'Content-Type': 'application/json' }, body: JSON.stringify({ fullName: formData.name, email: formData.email, college: formData.college, year: formData.year, phone: formData.phone, utmSource: new URLSearchParams(window.location.search).get('utm_source') || 'campus_landing' }) }); const data = await response.json(); if (data.success) { alert('Application Submitted Successfully! Check your email.')} } catch (err) { console.error('Submission Error:', err); } }
```

05 Step-by-Step Implementation Roadmap

| PHASE                       | TASK / ACTION ITEMS                                                                          | TOOLS & STACK                   | ESTIMATED TIMELINE |
|-----------------------------|----------------------------------------------------------------------------------------------|---------------------------------|--------------------|
| Phase 1: Design & UI        | Create Figma components matching NextGenGrowth Dark/Aurora theme & Campus Calculator layout. | Figma, Tailwind CSS             | 1-2 Days           |
| Phase 2: Frontend Dev       | Build responsive components (Hero, Calculator, Solutions, Network, CTA, Modals).             | React 18 + Vite + Framer Motion | 2-3 Days           |
| Phase 3: API & DB Hook      | Connect signup form to NextGenGrowth API endpoint & setup Resend auto-emails.                | Express API + MongoDB + Resend  | 1 Day              |
| Phase 4: Analytics & Launch | Set up Meta Pixel, GA4, test mobile responsiveness, deploy on Vercel / Cloudflare Pages.     | Vercel / Cloudflare, GA4        | 1 Day              |